

PORTFOLIO

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Introduction of an **urban perspective** to produce **green spaces** managed by the private sector by activating the carpark, green elements, and **enforcing identity**

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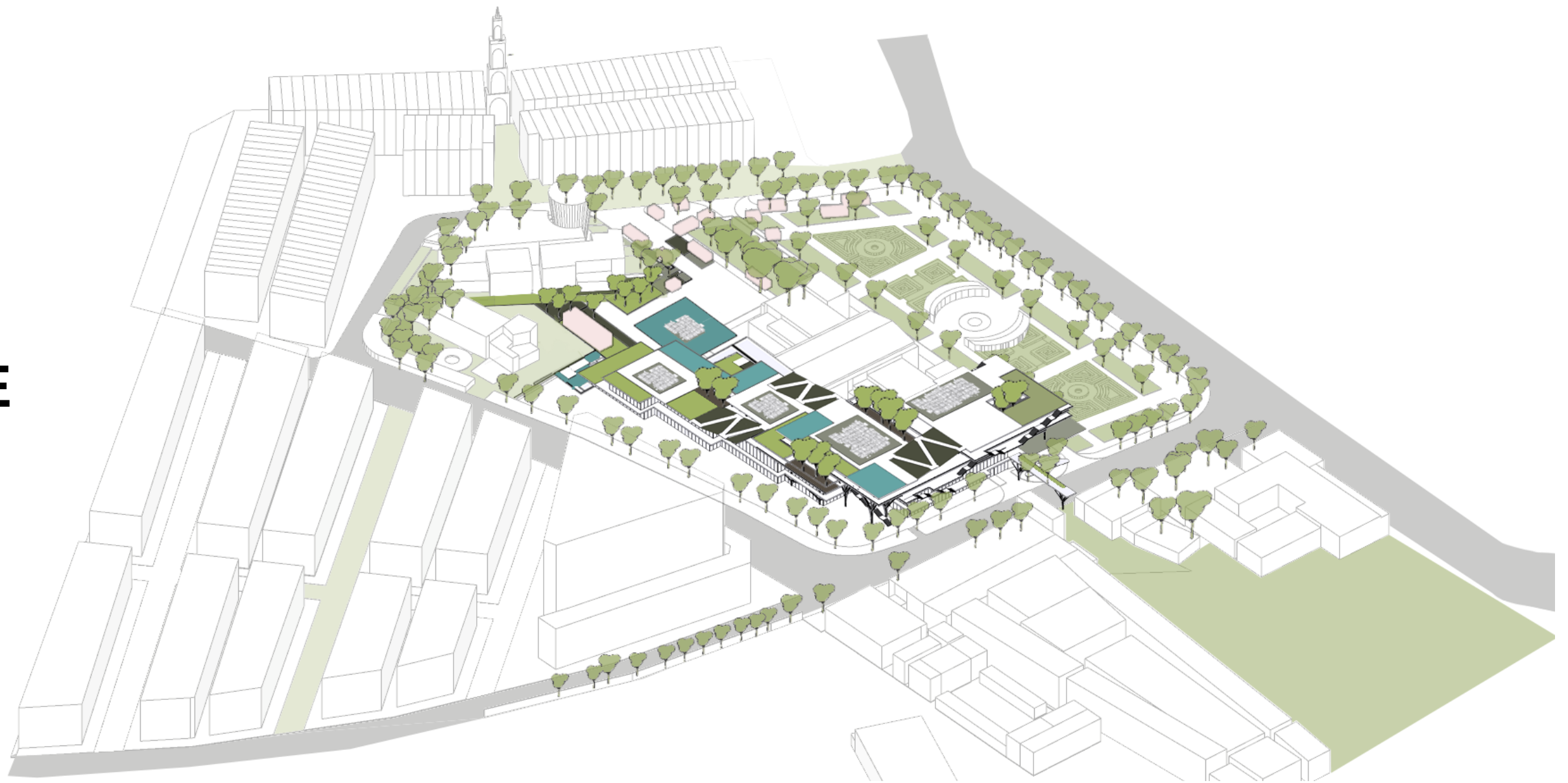
A challenge to build residential project to connects **human** with **nature** on an acquired land in a dense **urbanized area** without the **loss of culture**

03 OTHER EXPERIMENT

A **journey** of other courses, showcasing the **progress** and **development** in the academic years

THE JUNGLE

From City in the Jungle
To Jungle in the City



Comprehensive Analysis

The satellite cities in Phnom Penh are normally the consequences of the investment private sector resulting in what the other researches called "**inaccessible green spaces**".

Koh Pich is an expansion of Phnom Penh as a satellite city in which the prevalent interest is related to the margin of profit of a company.

At present, the area is **fragmented** by roads. However, there is sufficient green to potentially be connected a network within the island.

To become the **central hub**, the City Hall area would need to be connected to the Fair, the Elysee and the rest of the island.

The analysis envisioned not only the **green site**, but also the **road structure** connecting the **green network** to existing buildings.

- Blue Corridor
- Potential Green Corridor
- Site Boundary
- Primary Road
- Secondary Road
- Tertiary Road
- Nodes
- Potential Extension Building
- City Hall Area
- GreenArea
- Green Corridor



Concept Diagram



Site

The site is located in portion of the area where the green elements are fragmented and enclosed.



Extrusion

The mass is split and extruded to create variation of heights connecting the ground level up to the height of the city hall.



Offset

Then it is offset to reduce to perception of having a massive object in the context. In addition, the green terraces are introduced on top of the building.



Subtraction

The corners are being subtracted to make welcoming connections to the public rather than having a solid and formal object opposing to the street.



De-massing & Characterization

The visual impact of the building on the site is further reduced by treating the floors of the building as superimposed layers and avoiding the use of vertical facade windows.



Contextualization

The landscape offers a seamless uninterrupted experience connecting the shophouses, to the Town Hall, then through the building to the other side and along the main road to the

Masterplan

The city hall is located in an **isolated area that is disproportionate to its surroundings and isolated by a fence**. However, there is adequate green infrastructure and potential to create a green corridor towards the river, thus forming a **network of green spaces across the island**.

The building will offer **rooftop spaces and a bridge** connecting to Treelion Park, the river, and other nearby buildings. This will establish the building as a central hub on the island, providing **accessible public green areas and convenient pathways**.

The building is designed to **attract young people and is activated through its various functions** , such as an open market, vendors, shops, anchor shops, sports, offices, cafe and bar, which are **integrated into the green space and the parking lot**. The building itself would also allow for flexible spaces for events and promotions. The diverse range of activities available to the public on site would **activate the green space and add value to the surrounding area**.

- N
1.

Street Vendors
2.

Urban Forests
3.

Market/ Multi-purpose Area
4.

Restaurant
5.

Cafe
6.

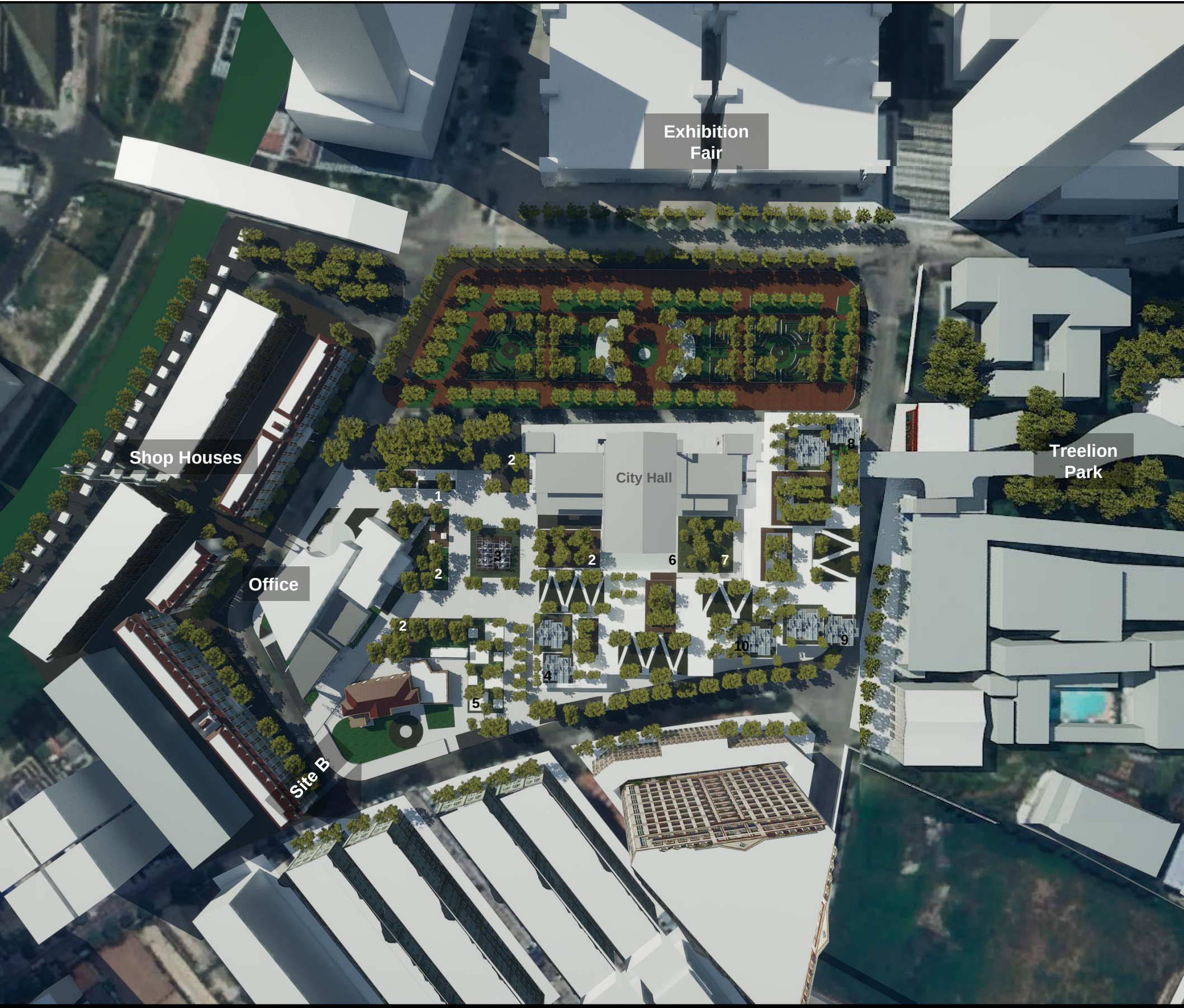
Wedding Platform
7.

Exhibition Area
8.

Bar
9.

Pool
10.

Promotion Area



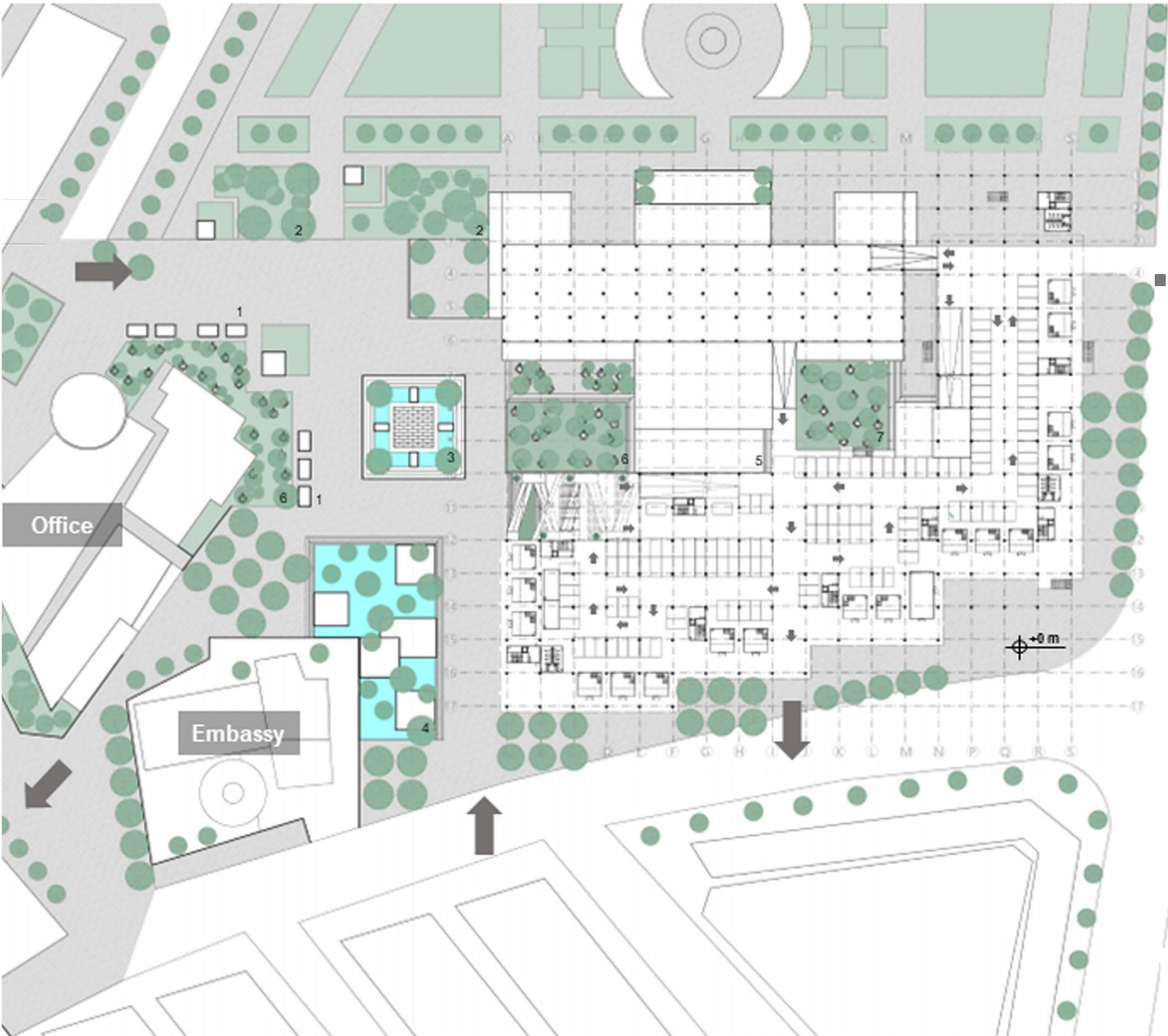
Ground Floor

On the ground level, from the main entrance, there are small **subdivisions of steps** which are functioned as vendors as to welcome, visitors. Passing through small **terraces and vendors**, the landscape invites the users into the area with a **formal element** at the west wing of the city hall and the market at the south.

From this point onwards, **the spaces are decentralized from the pavilion** towards different directions. Directing towards southwest is a connection to another site with a journey through **urban forests**.

On the south side of the market, is the cafe with seating places being ventilated, shaded, floated on the water, giving the experience of **submerging into the water and inside a forest**.

Another direction towards the east, going through the urban forests, in a series of steps across the wedding platform and to the other side of the city hall, offering the **sensation of terraces and jungle**.



- N

⌚
- 1. Street Vendors
 - 2. Urban Forests
 - 3. Market/ Multi-purpose Area
 - 4. Cafe
 - 5. Wedding Platform
 - 6. Outdoor Cafe
 - 7. Exhibition Area

Renderings



Arial View, Southeast Corner: The elevated car parks and shops are connected with stairs to the rooftop. Then the city hall's park introduces green canopies, connecting to the food street at the water channel.

Renderings



Arial View, Southwest Corner : The green element is introduced and continues upwards with the sloping garden, the pavilion, sunken gardens as well as green canopies and recreation area.



Landscape: The landscape can take on a linear form, which expresses horizontality through the progression of steps leading to green and blue spaces.



Open Office : Office spaces are integrated into the landscape, enclosing only private spaces. The use of branching column, recessed back imposing only the horizontal slabs.

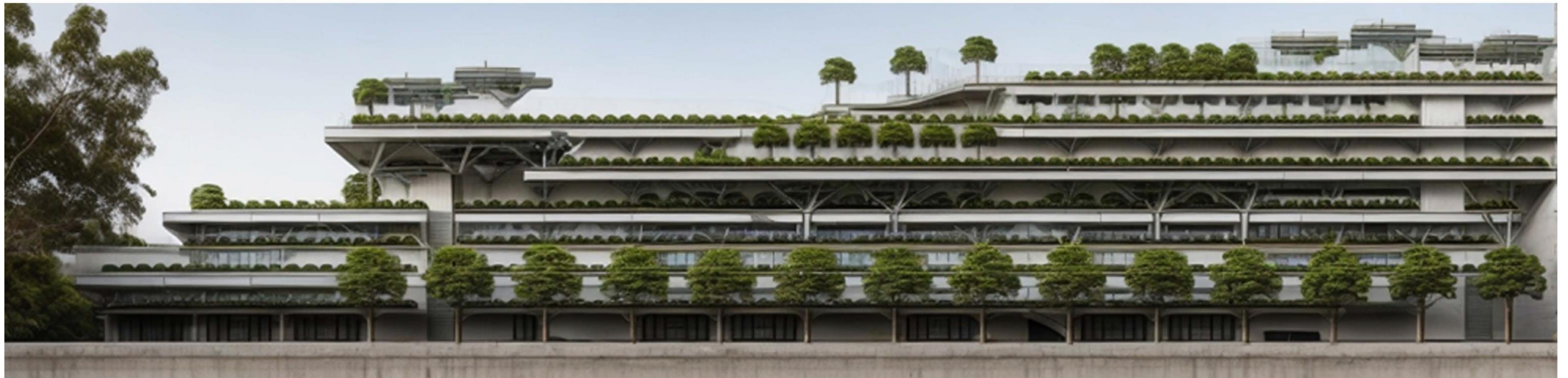


Gym : The emphasizes of natural element and horizontality of the spaces towards the interior. Introducing also the function such as gym and library to compliment with the office.

Renderings



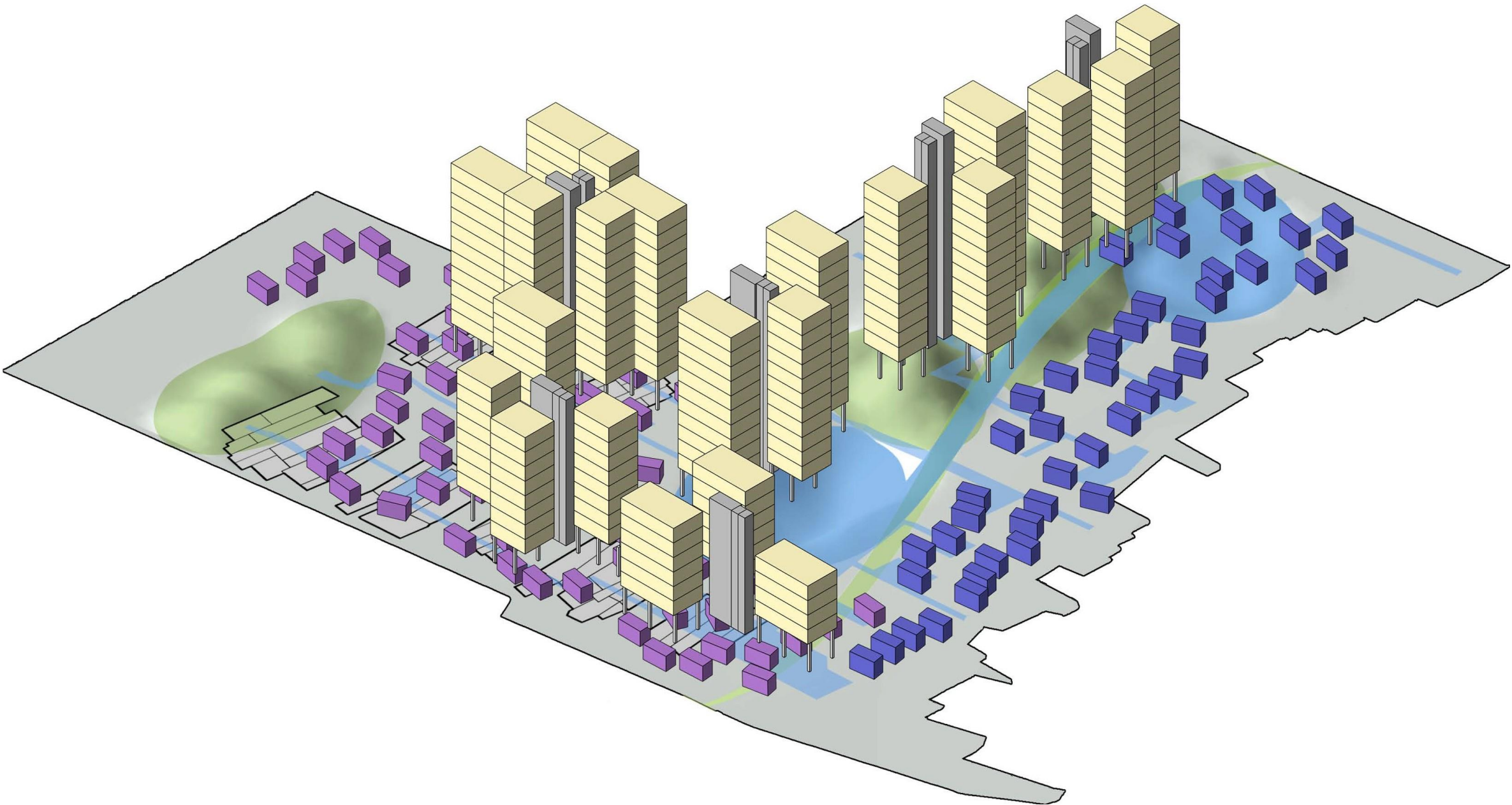
South Elevation: Garden Ramp, connecting to different levels of the terrace, along side the Sunken Garden and pavillions.



East Elevation: Superimposed Horizontal slabs, with recessed columns at the back, with Pavillions and Recreation spaces on the rooftops.

BIO-CITY

Restoration of the Cultural & Natural Connection in the City

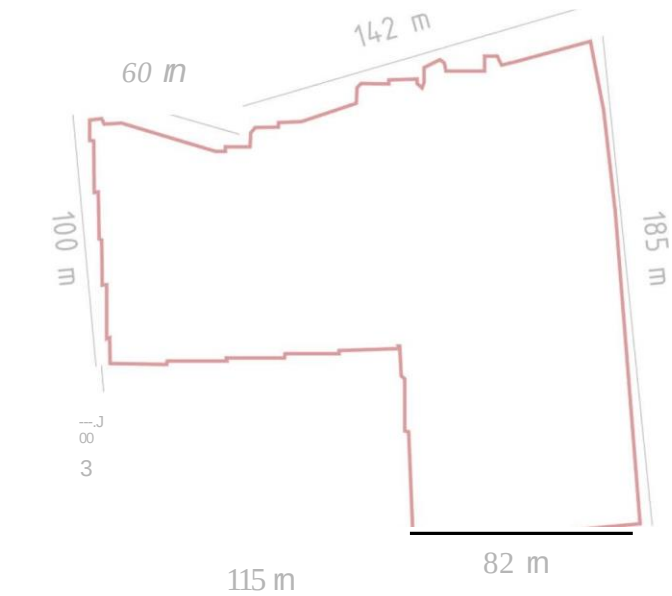


Location

Bio-city is a project that attempts to solve the problem of **dense housing in Phnom Penh**, Cambodia. In exchange for the development project, **a lake (Beoung Kok) was filled** to prepare for new high-rise building.

"**Bio**" refers to the life including the **life of the natural landscape**, and the **life of the people** who lives as part of nature as well. While "**City**" refers to the **habitat of the human being**.

The memories of the nature, such as the lake and the forest, and the identity of the people, lifestyle, culture, especially with Cambodians where water was once cherished by the civilization and has its whole culture around the water. This has been lost, blurred with the new idea of modernity and contemporary building.

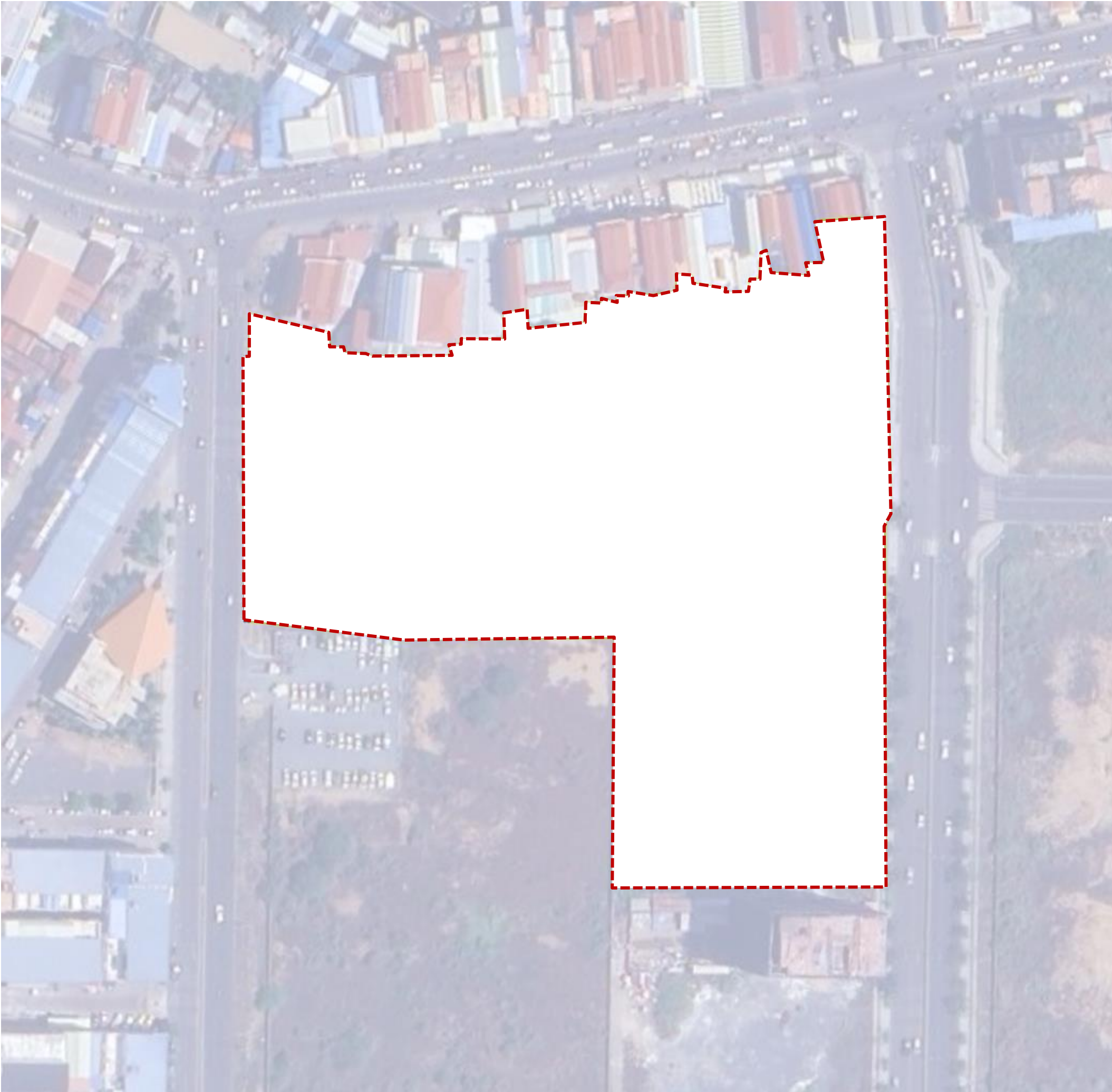


Area: 24500 sqm

Parameter: 762m



Site



Conceptual Form

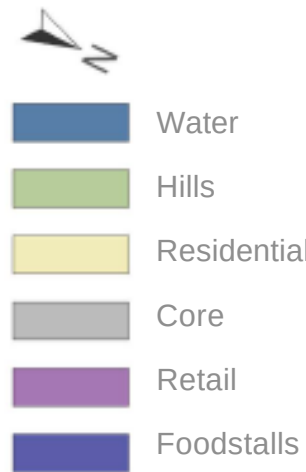
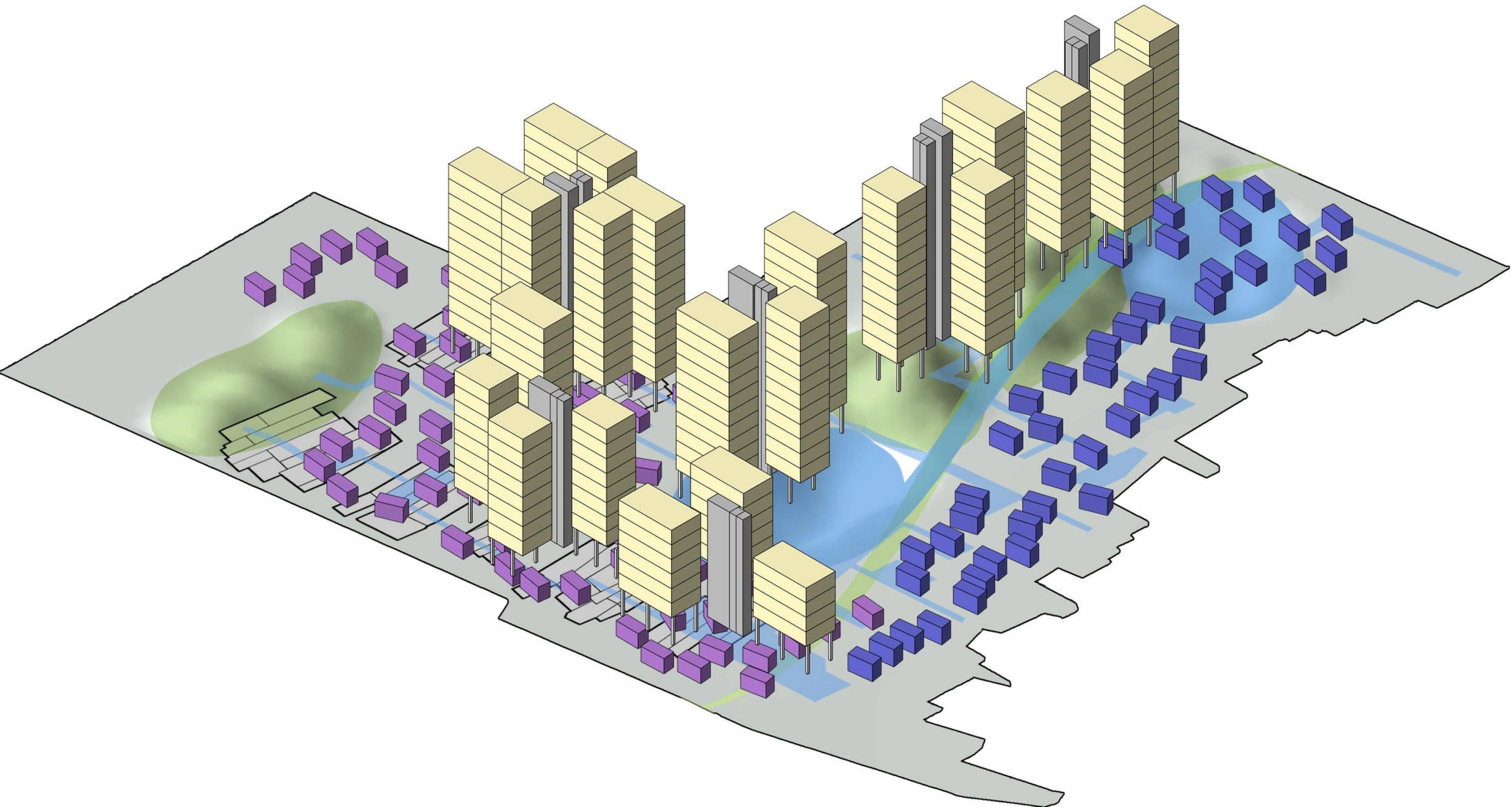
Bio-city firstly **respects the elements of nature that once existed in the site and re-imagine it.** The feeling of the lake is re-created along its axis, the **dug-up soil is refilled as hills, creating a natural landscape.**

Then the chaotic arrangement from the unplanned development at the north is respected, with its unique pattern, extending from the surrounding shop houses inwards the site, and **re-envision the patterns and houses in a new urbanized context.**

The patterns on the ground takes the shape of the **neighboring urban fabric** on top of the **linear water corridor for recollection.**

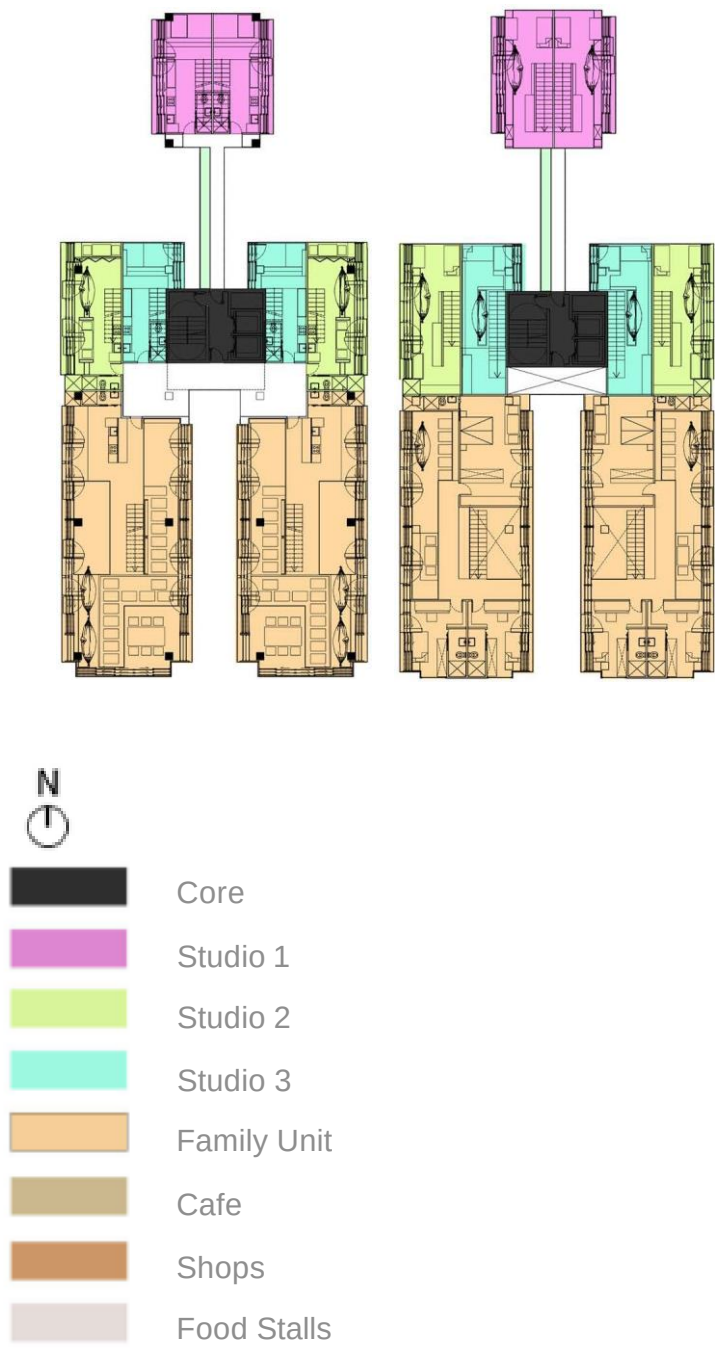
The **recreation spaces are arranged in the hills and lakes naturally** while the residential spaces are kept free from the ground level.

Other activities are also included to further enhance the sensation of living in a **traditional village that once existed in the lake.**



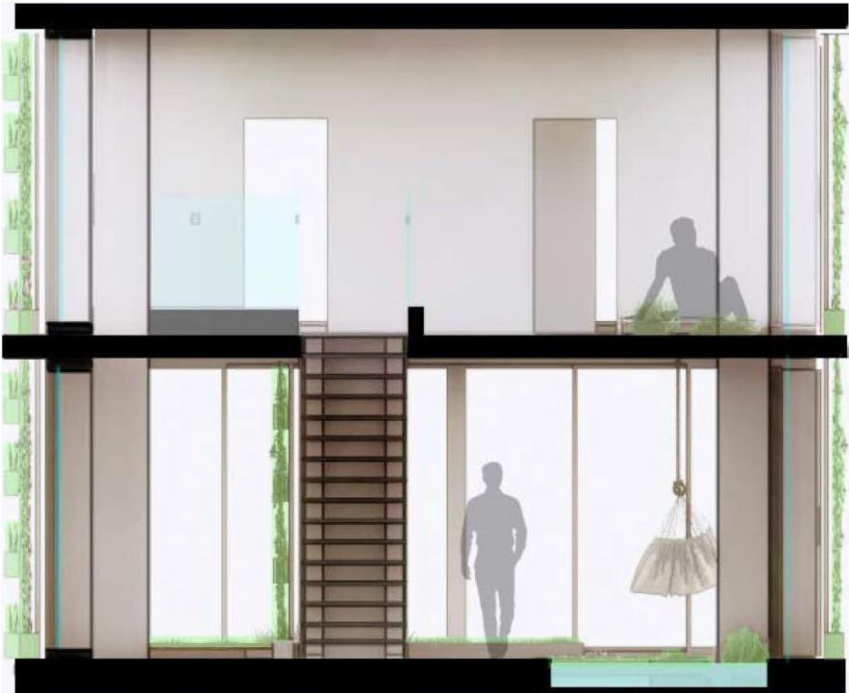
Floor Plans

The Shops, Cafe and Food Stall allow for activation of the ground floor. The building consist of **the core in the middle of 4 unit**, 3 of which are studio and a family unit. The arrangement of the building **allows for ventilation** from the prominent North wind, while keeping **the east and west side shaded** from the sunlight.

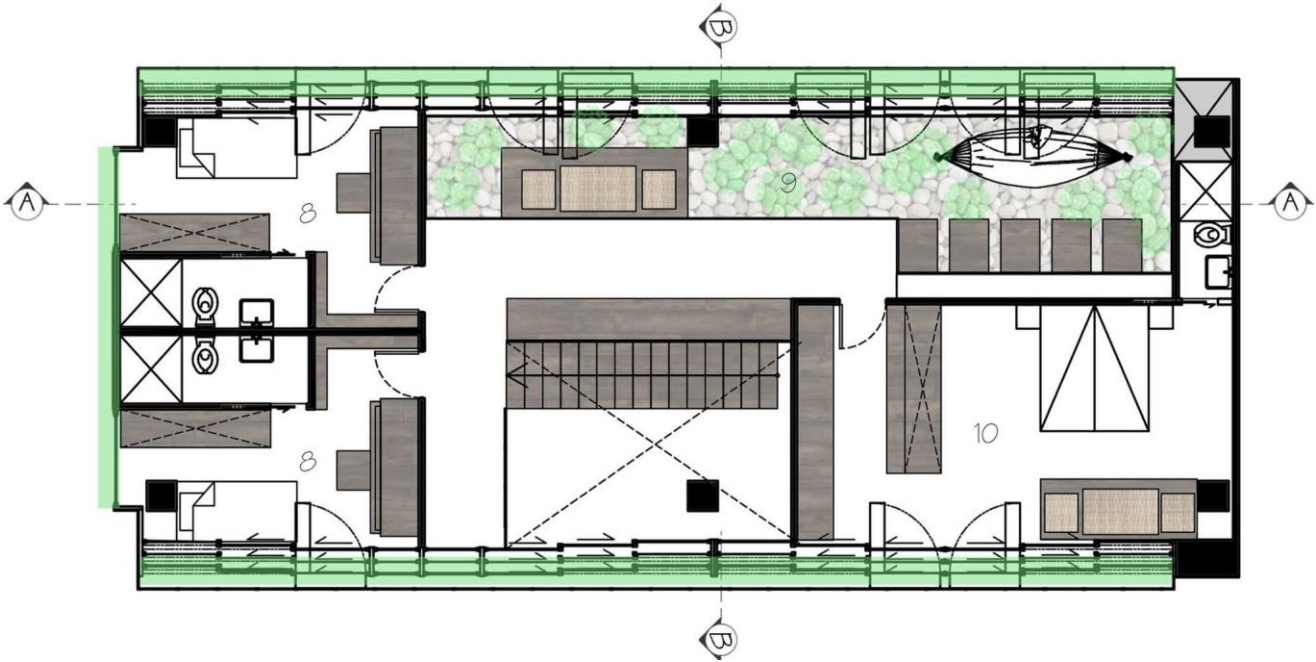
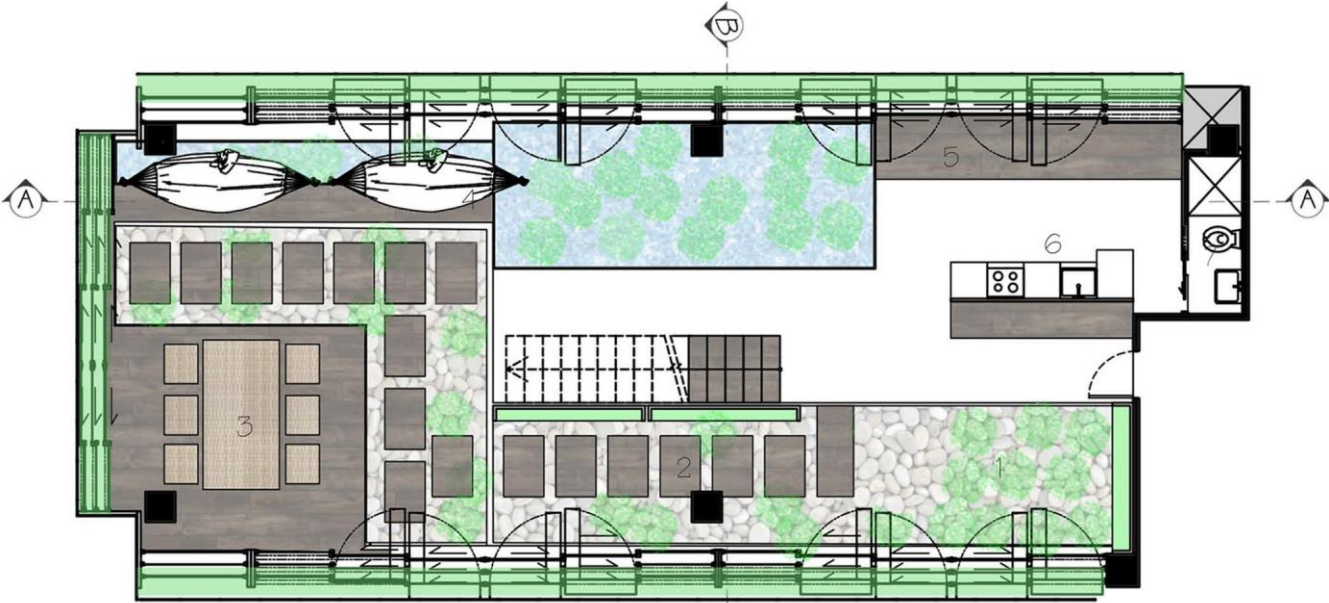


Family Unit

The Family Unit embraces the idea of **cultivating green vegetation** allowing for natural ventilation, sunlight and the experience of **emmersion with the nature while living at home**. The **water element and green spaces** are also included in the unit along side the usage of **the wood and hammock as memories of the traditional houses**.



Section B



Section A



- | | | | |
|---|----------------------|----|----------------|
| 1 | Garden | 6 | Kitchen |
| 2 | Green Corridor | 7 | Toilet |
| 3 | Dining / Living Area | 8 | Single Bedroom |
| 4 | Relaxing Area | 9 | Relaxing Area |
| 5 | Maintainance Area | 10 | Master Bedroom |

Renderings



Green Corridor: Activating the ground level with green corridor, vendors, and other activities while the above floors remain as residential units.



Interior : Inviting the natural element to the space, recreate the family relationship in a traditional lifestyle while adapting to the urbanized context

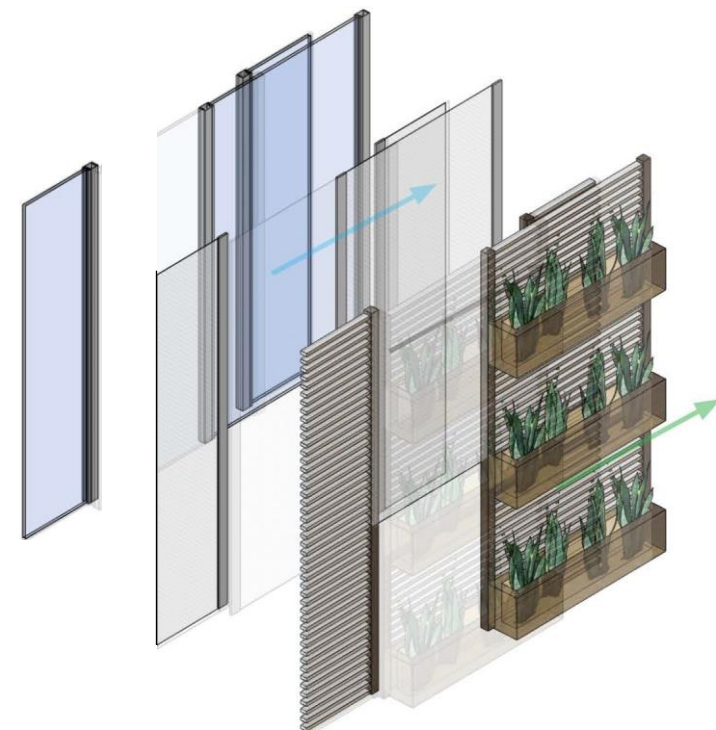


Interior : Harmonious usage of traditional elements and the landscape that once existed around the site, including earth, water and wood.

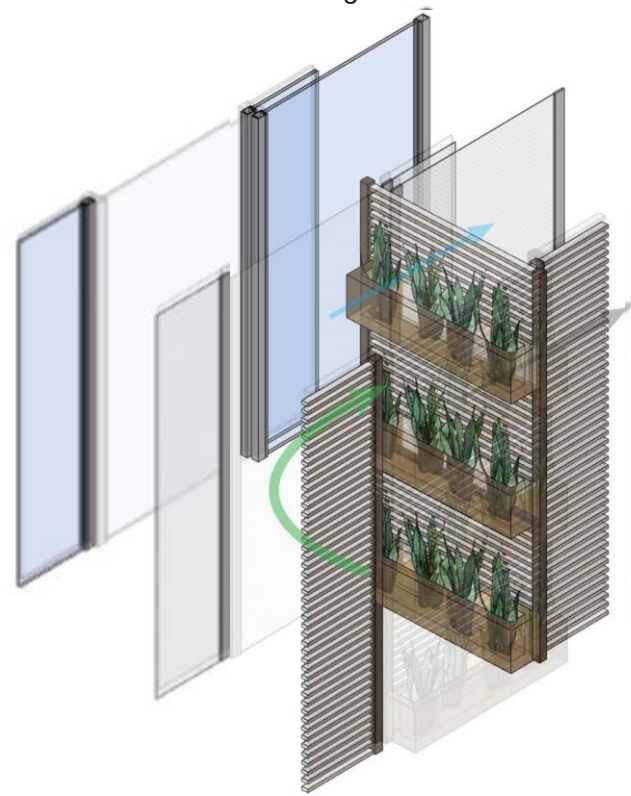


Interior : Unifying vertical farming, natural landscape and Khmer lifestyle together to express modernized Khmer culture while still being sustainable.

Facade Details



Sliding



Pivoting



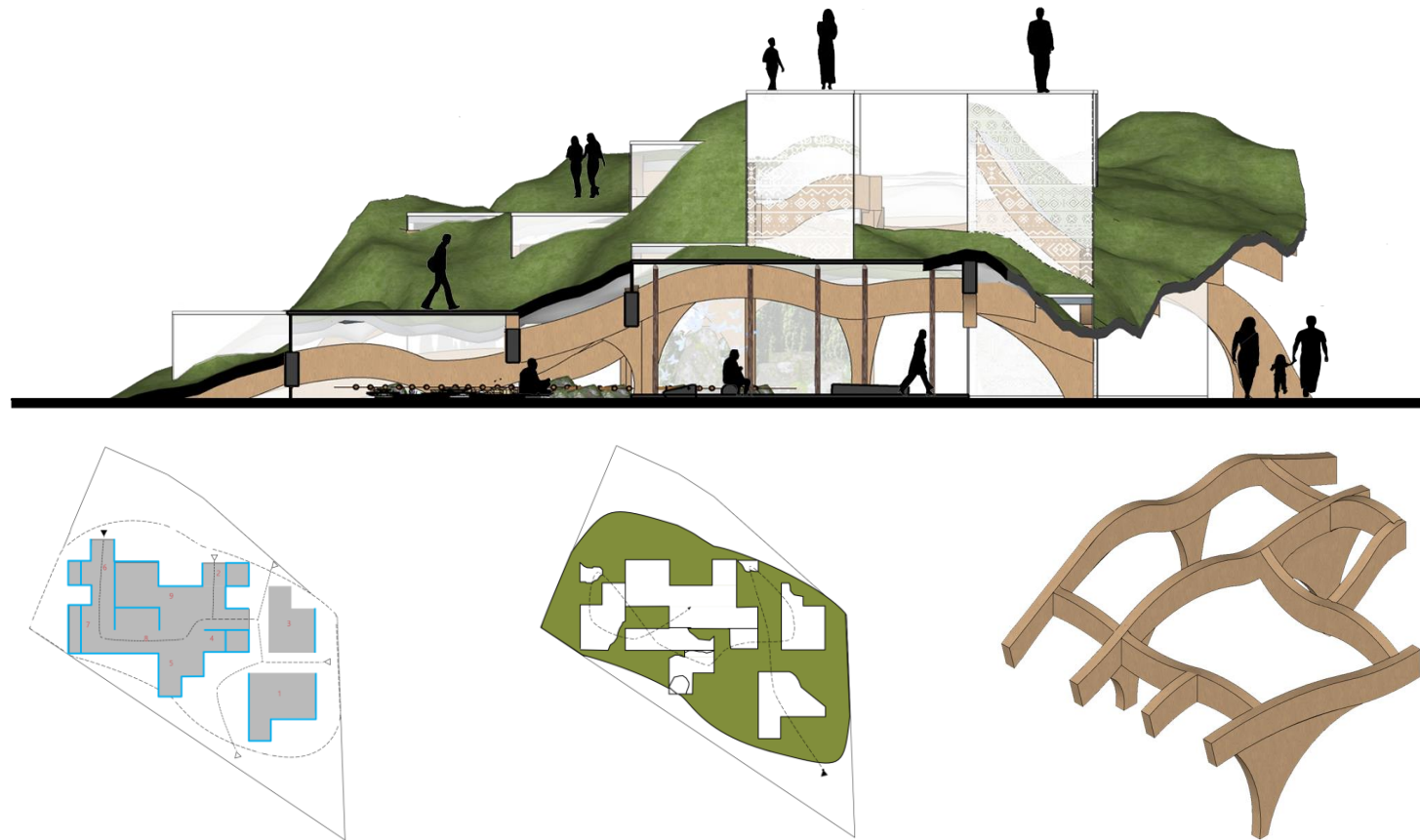
Facade Detail : The facade system allows each panel to be stationary, slided, or pivoted to maintain the vegetation. Variaion of facade panels also make it possible to create a constantly changing elevation in which each panel.

OTHER EXPERIMENTS

Exploration of various
design approaches



The 9 Paradise



Design Process : The 9 Paradise was meant to be a competition project for Slovak Exhibition Expo in Japan, Showcasing the territory and culture of Slovakia.



Cave : A small sitting area with lights mimicking the spatial sensation of a Cave, locating in a low height space, and directional light from above.

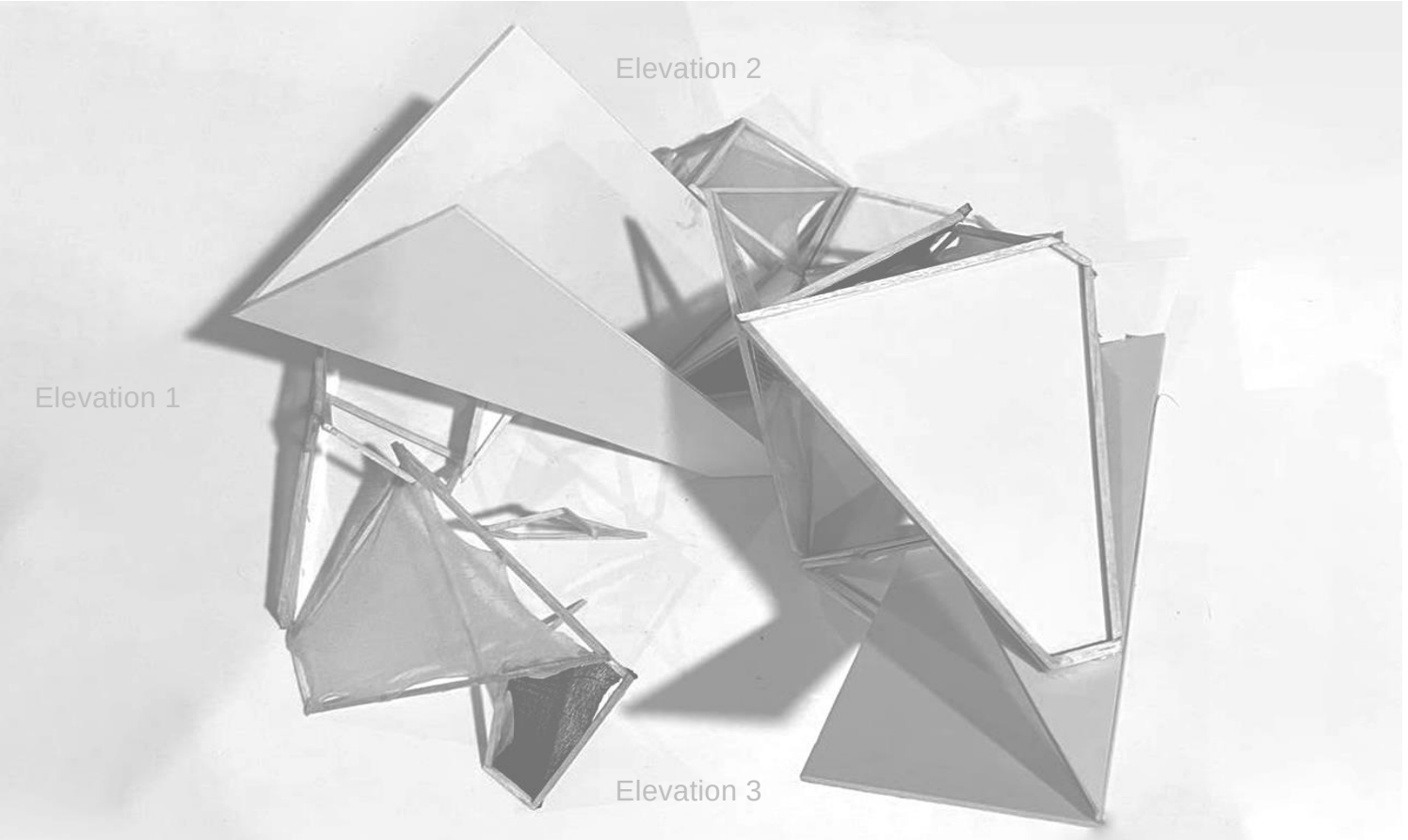


Exhibition Space : A double-height space, supported by glasses and curved columns, viewing the rich cultural properties of patterns on the glass and its terrains outside, connecting the interior and exterior.



Forest : The sensation of a forest is brought through a big space supported by tree-inspired columns. The soul and logs of trees with low light settings, delivers the experience of being in a forest.

Picasso's Viewing Gallery



Top View : Picasso's Viewing Gallery is an **experiment of form following** the concept of cubism. It takes 3 painting of **"The Accordionist"** of Picasso and divided it from different perception of the Accordionist.



Elevation 1 :The material is further adapted to show **the difference of elements**. This includes the use of Solid, Reflective, Transparent, and Semi-Transparent.

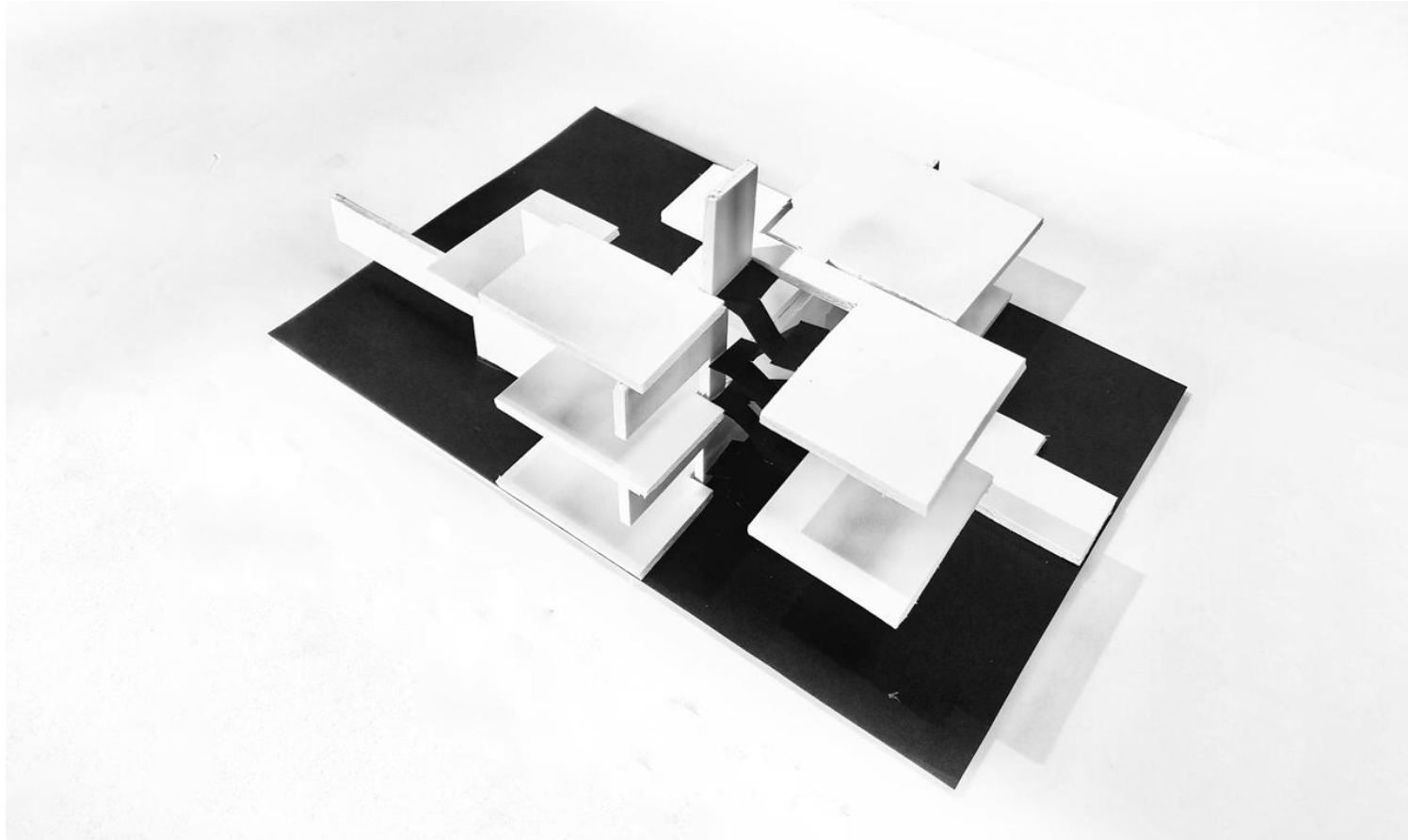


Elevation 2 : The structure is also adapted to the different material, making is as a **solid-load bearing structure, frame, and tensile structure**.

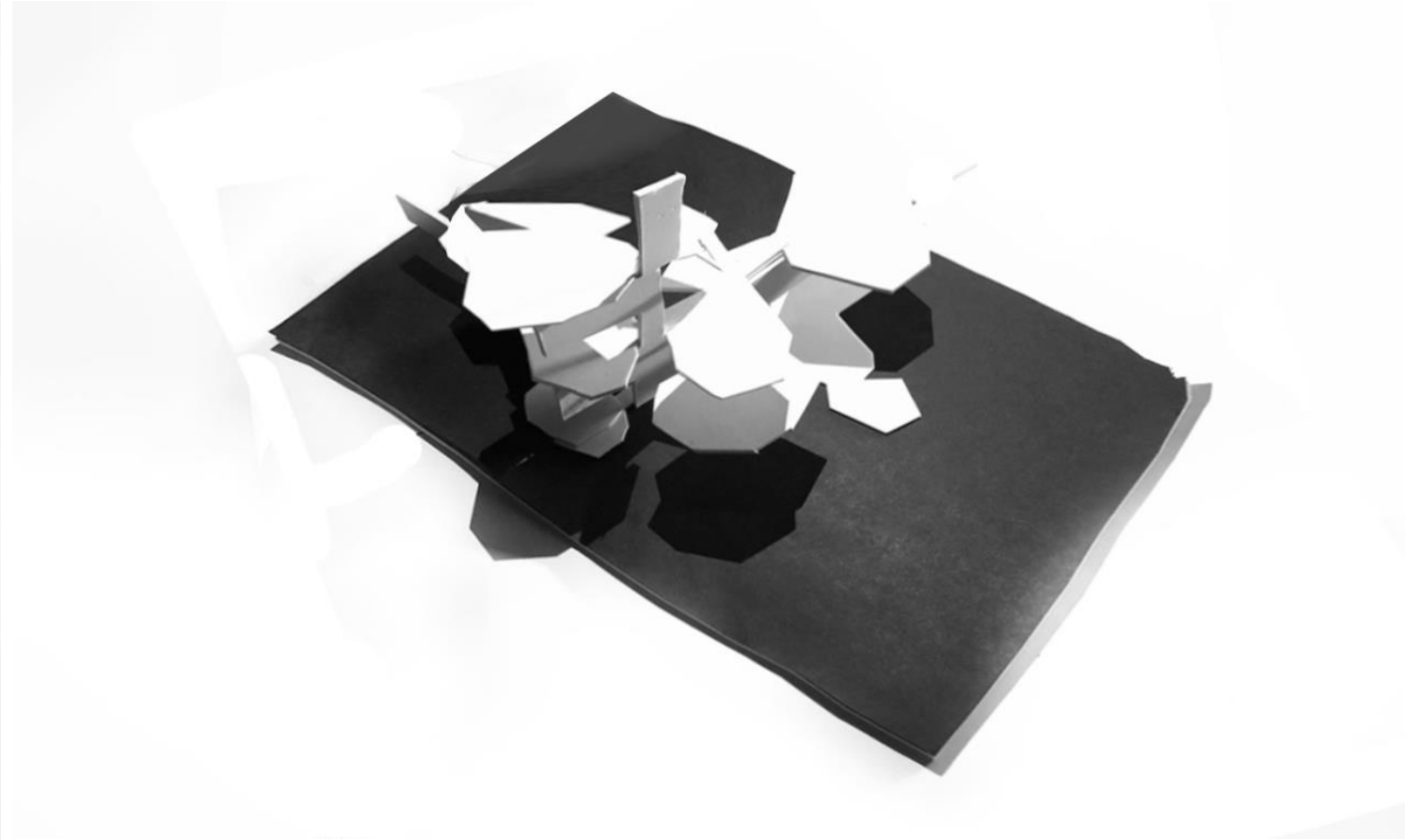


Elevation 3 : The resulted product is a statement of strong element in the center, tilted outwards, supported by light elements. As one walk through the space, they will experience **the changes of light, shadow, reflection, solidity etc.** as the property of material differs,

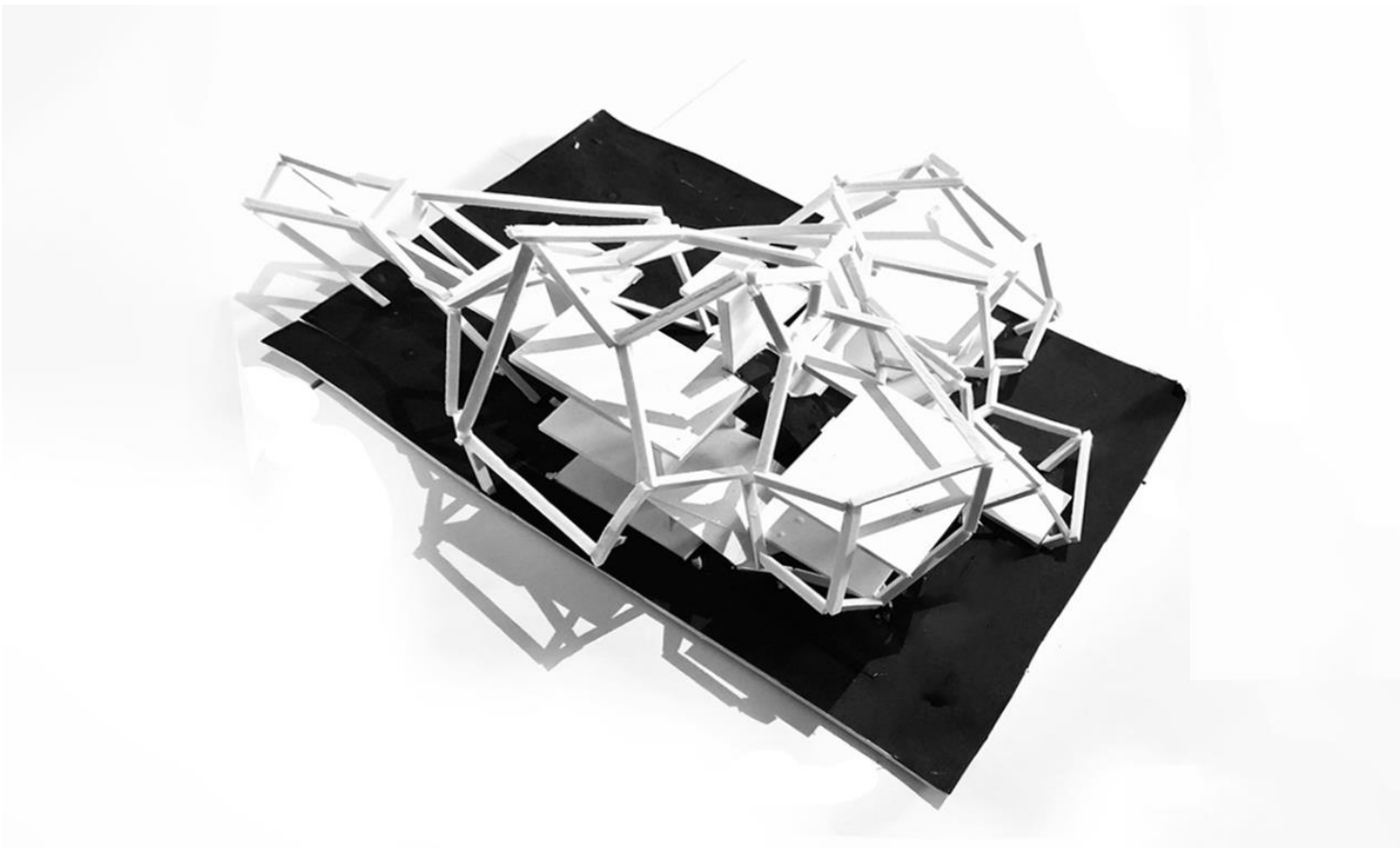
Geometric, Organic Shape & Biomimicry



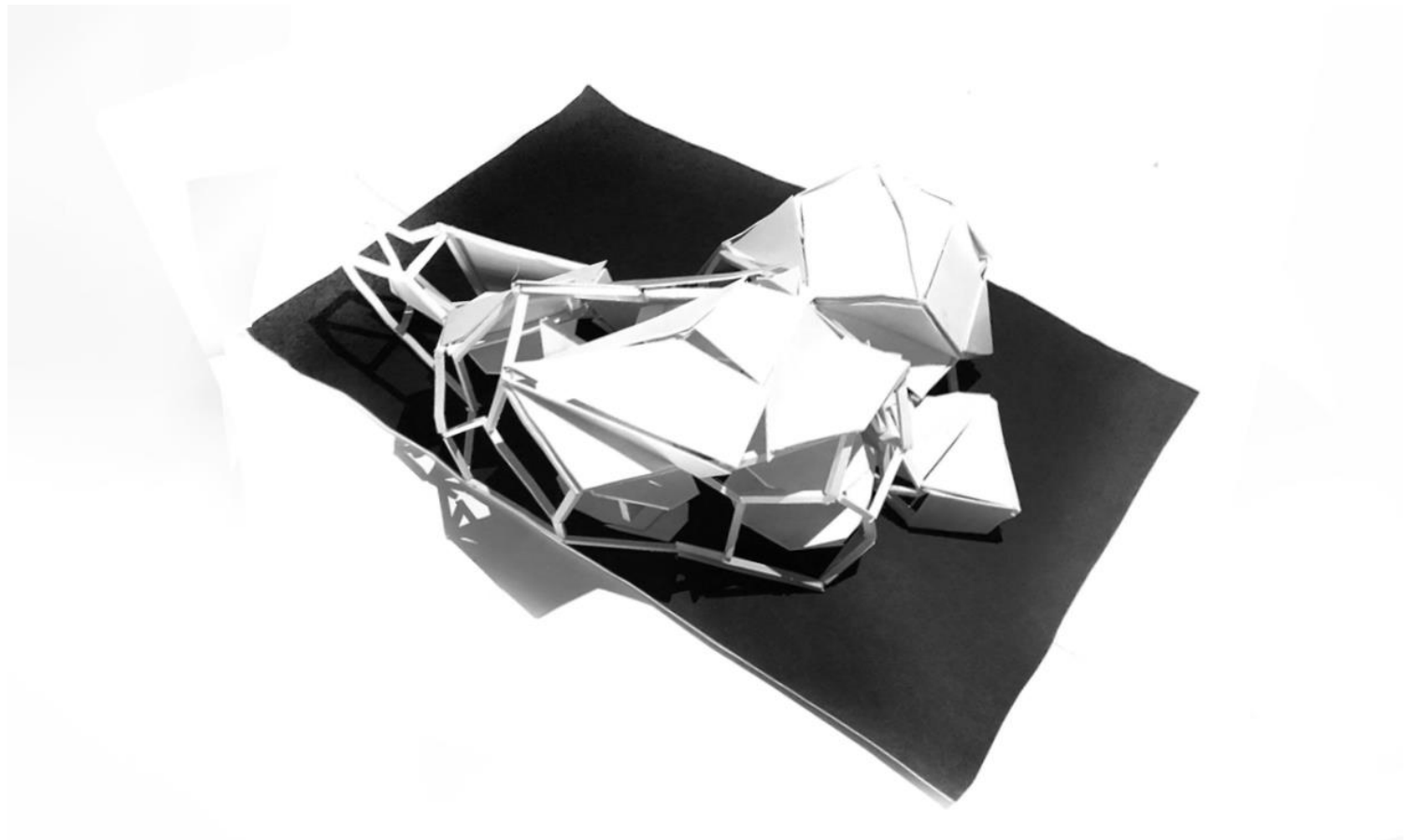
Composition of Geometric Shape: The spaces are arranged with **different composition on each level**, and staircase in the middle. A horizontal element is protruded, exaggerated to make the shape dramatic.



Adjusting the Geometric shape : The rectilinear shape of the levels are subtracted to **mimic the shape of the crystal**.



Formal Expression : The Crystal is further studied to **give the form to the levels**.



Opening and Enclosure : Panels are then added to **create the opening and enclosed spaces**, while keeping both the interior and exterior the same language of geometry..